T1D EDUCATION | STORY NO. 3

Type 1 vs. Type 2 Diabetes

Busting the Biggest Myths - starting with how a healthy cell works

By Aviva

"Did you get diabetes from eating too many sweets?"

No. That is the most common question people ask, and sweets did not cause Type 1. Read the diagrams in this handout in order - healthy first, then Type 1, then Type 2, then gestational - and you can see exactly where each one is different. Same cell, same door. What changes is the key.

How a healthy cell works



1. Pancreas
Beta cells inside it make insulin

→



2. Insulin = a key
Insulin travels in the blood - the key turns easily

→



3. Cell door OPENS
The key turns easily and unlocks the cell. Glucose goes inside

1. **You eat or drink.** Food is broken down into a sugar called glucose. Glucose travels around the body in the blood, looking for cells that need energy.
2. **The pancreas makes insulin.** Inside the pancreas are tiny beta cells. When blood sugar rises after food, the beta cells release insulin into the blood.
3. **Insulin is the key - and it turns easily.** Insulin reaches a cell, fits into a lock on the cell's surface, and turns easily, like a good key in a good lock.
4. **The door opens and glucose goes in.** Once the cell is unlocked, glucose moves out of the blood and into the cell.
5. **The cell makes energy.** Inside, glucose is used as fuel - to think, move, play and grow. Blood sugar comes back down to a healthy range.

Healthy result: insulin is made, the key turns easily, the door opens, and glucose becomes energy - not a pile-up in the blood.

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Type 1 Diabetes - what goes wrong: no key at all



1. Pancreas ATTACKED
Immune system destroys the beta cells

→



2. No key at all
No insulin is made, so there is no key at all

→



3. Cell door stays SHUT
Glucose cannot get in. It piles up in the blood

1. **The immune system makes a mistake.** Type 1 is an autoimmune condition. The body's own immune system, which should fight germs, accidentally attacks the insulin-producing beta cells in the pancreas.
2. **The beta cells are destroyed.** As they are lost, the pancreas makes little or no insulin - the beta cells that should make the keys are gone.
3. **With no insulin, there is no key at all.** Glucose arrives at the cell, but nothing unlocks the door.
4. **The cell stays shut and goes hungry,** while glucose builds up in the blood instead. That is high blood sugar.
5. **Insulin has to come from outside the body** - from a pump or injections - every day. Nothing I ate or did caused Type 1, and it cannot be cured by diet or exercise.

Type 1 in one line: no key at all - external insulin is needed, by pump or injections, every day.

Type 2 Diabetes - what goes wrong: the key turns halfway, then gets stuck



1. Pancreas still works
Beta cells still make insulin and keys

→



2. Key turns halfway, then gets stuck
The key is there, but the lock is stiff and rusty

→



3. Cell door barely opens
Only a little glucose gets in. The rest stays in the blood

1. **The pancreas still makes insulin.** The beta cells are not attacked and destroyed the way they are in Type 1.
2. **But the cells stop listening to it.** Type 2 is a metabolic condition where the body becomes resistant to insulin, or does not produce enough of it over time.
3. **The key turns halfway, then gets stuck.** The lock is stiff and rusty, so the door barely opens.
4. **So glucose builds up in the blood too,** but for a completely different reason than in Type 1: the key exists, it just turns halfway, then gets stuck.

Type 2 in one line: the key is made, but it turns halfway, then gets stuck - that is insulin resistance. Different colours, different problem.

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And gestational diabetes is different again

Gestational diabetes only develops **during pregnancy**. It is not Type 1 and it is not Type 2.



1. Pancreas still works
Beta cells still make insulin and keys

→



2. Key is difficult to turn
Placenta hormones make the key difficult to turn

→



3. Cell door opens less
Less glucose gets in, so blood sugar rises in pregnancy

- **What happens:** hormones from the placenta make the mother's insulin harder to use - the key exists, but placenta hormones make the key difficult to turn - a temporary form of insulin resistance, a bit like the stuck lock in Type 2.
- **What is different:** the immune system does NOT attack the beta cells, so it is not autoimmune like Type 1. It starts during pregnancy, mother and baby are monitored closely while she is pregnant, and blood sugar **usually goes back to normal after the baby is born**.
- **Why it matters here:** a pregnant adult with gestational diabetes is not a kid with Type 1. Different cause, different timing, different care - so please do not mix them up.

Why gestational diabetes is easier than Type 1

In gestational diabetes, the key still exists. The pancreas still makes insulin - in fact it makes more during pregnancy - but placenta hormones make that key difficult to turn. In Type 1, there is no key at all: the beta cells are destroyed, so insulin must come from outside the body every day, for life. Gestational diabetes is also temporary - when the baby and placenta are delivered, the cause goes away and blood sugar usually returns to normal. That is why the effort is lower: often food choices, activity and blood-sugar checks for a few months, though some people need insulin. Type 1 never pauses and does not go away.

Gestational in one line: the key is difficult to turn - placenta hormones make it stiff - and it usually ends after the baby is born.

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At a glance



Healthy
Key turns easily, door opens

→



Gestational
Key is difficult to turn

→



Type 2
Key turns halfway, then gets stuck

→



Type 1
No key at all - external insulin needed

Same word - diabetes - four very different jobs. The last column shows how much work each one takes, and how disruptive it is.

	Is insulin made?	Does the key work?	What starts it?	Effort to manage - and how disruptive it is
Healthy	Yes	Key turns easily - door opens	-	**None.** Automatic - the pancreas does it all. No counting, no monitoring, no insulin.
Gestational	Yes	Key is difficult to turn - placenta hormones	Placenta hormones in pregnancy	**Low to medium.** Often managed with diet, activity and blood-sugar checks during pregnancy; some need insulin. Usually ends after birth.
Type 2	Yes, at first	Key turns halfway, then gets stuck	Insulin resistance (metabolic)	**Medium to high.** Lifelong, but often managed at first with food, activity and pills. Some people later need injections or insulin.
Type 1	Little or none	No key at all - external insulin needed (pump or injections)	Autoimmune attack on beta cells	**Hardest - highest.** Lifelong and extremely disruptive: insulin every day by pump or injections, CGM monitoring 24/7, carb counting at every meal, and daily ups and downs.

Understanding the difference helps eliminate stigma and ensures kids with Type 1 get the exact medical support and understanding they need at school and sports. Same word - diabetes - but different causes, so the support has to fit the condition.

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Case study: one cup of boba tea - three very different reactions

Same cup, same sweet tea, same pearls. A healthy body handles it automatically: the pancreas releases just enough insulin and blood sugar settles back down. For diabetes, what happens next depends entirely on the key - and that is why one drink is a small thing for one person and a whole project for another.



The sweet tea hits first. Sugary liquid is digested fast, so glucose races into the blood within minutes.

The pearls hit later. Tapioca pearls are pure starch. They are chewed slowly and digested over a much longer time - a second wave of carbs after the first.

The total is big, and hidden. One large sweet cup can carry as many carbs as a whole meal, and the syrup coating the pearls adds sugar you cannot see.

Type 1 - no key at all, external insulin needed, and the whole cup is a guess

Her pancreas makes little or no insulin, so nothing in her body will slow this rise on its own. Every gram in the cup has to be counted - guessed, really - and matched with insulin from her pump or injections, following her own care plan, usually with an adult's help. If the guess is wrong, the sugar wins. This is the hardest reaction of the three.

Type 2 - key turns halfway, then gets stuck

The pancreas still makes insulin, so the rise is not unopposed - but insulin resistance means the glucose clears slowly, so blood sugar goes higher and stays high for longer. There is usually no minute-by-minute matching to do for one drink. The problem builds differently: sugary drinks make an already hard job harder, again and again, over time.

Gestational diabetes - key is difficult to turn

A pregnant adult still makes insulin - in fact, more than usual - but placenta hormones make it work less well, so a very sugary drink pushes blood sugar up more than it normally would. She and the baby are monitored, and she usually follows a food plan during pregnancy to keep levels steady. It is real work, but it is temporary, and unlike the Type 1 child, she has a working key the whole time.

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